🛠️ TECHNICAL SPECIFICATIONS: ANALYTIQ PRO (V.1.4)

Strategic Vision: AI-Agentic Solution Architecture for Business Valuation

Core Principles: Transparency, Human-in-the-loop Governance, and Materiality.

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1. ARCHITECTURAL DECISION RECORD (ADR) - REVISED

Date	Decision	Rationale (Justification)
31/03/26	**Agentic Orchestration: LangGraph**	Transition to a controlled state-machine. **MVP Strategy:** Start with a **Linear Graph (A → B → C)** to manage the learning curve before complexity.
31/03/26	**Governance: The Transparency Ledger**	Every AI decision logged in SQLite. Audit focus on **Material** changes only.
31/03/26	**Strategic Pivot: SQLite-First**	Local-first development for rapid prototyping, GitHub portability, and zero-cost scaling.

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2. UPDATED TECHNICAL STACK (REFINED)

- **Orchestrator:** LangGraph (Open-source).
- **Intelligence:** Gemini 1.5 Pro/Flash (via Google AI Studio).
- **Database & ORM: SQLite** + **SQLAlchemy** (for clean Python-to-SQL mapping).
- **Data Validation: Pydantic** (to enforce strict schemas for incoming user data).
- **Backend:** FastAPI (Python).
- **Frontend:** React/Vue (Modern SaaS interface).

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3. AGENTIC WORKFLOW & GUARDRAILS

- **Skill 1: The Mapper (Ingestion Agent)**
  - *Task:* Semantic mapping to the 6-dimension Valuation Matrix.
  - *Output:* Lineage Note + Proposed Schema.
- **Skill 2: The Auditor (Cleaning Agent)**
  - *Task:* Logical & Chronological anomaly detection.
  - *Loop:* Propose-then-Execute with mandatory Human Gate.
- **Skill 3: The Quant (Calculation Agent)**
  - *Guardrail (Python-based):* Replaces second AI calls for cost efficiency. Includes:

        - Syntax verification.
        - Mandatory WHERE clause check.
        - **Strict prohibition** of DELETE/DROP commands without high-level admin override.

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4. DATA GOVERNANCE & MATERIALITY

- **The Transparency Ledger:** SQLite table tracking Timestamp, Action, Reasoning, SQL, and Approval.
- **Dynamic Materiality Threshold:** * **Default:** 5% impact on final KPIs.
    - **Adjustable:** Users can configure the threshold (1%, 5%, 10%) based on the business context (e.g., lower for high-precision audit, higher for quick valuation).
- **Cross-Column Logic:** Hard-coded Python rules (e.g., Order Date >= Signup Date) that act as an immutable truth before AI processing.

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5. GLOBAL & REGIONAL POSITIONING

- **Interface:** Arabic, French, English.
- **Outputs/Reports:** Strictly **English** (Standard for UAE & International job markets).
- **Personal Branding:** Positioned as an "AI Solution Architecture" project on GitHub/LinkedIn, demonstrating mastery of the AI Agent trend.

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6. ROADMAP: THE "LOCAL-FIRST" MVP

1. **Phase 1 (The Kernel):** Linear LangGraph script for Ingestion + Mapping + SQLite table creation.
2. **Phase 2 (The Audit):** Integration of the "Cleaning Dialogue" with human validation gates and adjustable materiality.
3. **Phase 3 (The Brain):** Full SQL mapping to the 6-dimension proprietary Matrix.
4. **Phase 4 (The Web UI):** Transition from CLI/Terminal to a full-stack React/FastAPI interface.